BASBO AYELAZONO

Basbo.Ayelazono@trojans.dsu.edu — 605 671 9329 — LinkedIn — basboayel.github.io

PROFESSIONAL SUMMARY

Interdisciplinary research student combining computer science, biology, and human-centered design to study assistive technology, computational biology, AI, and health-focused systems. Experienced in bioinformatics, experimental research, and accessibility-oriented device prototyping. Actively seeking PhD positions for Fall 2027.

EDUCATION

Dakota State University (Bachelor of Science) — Expected Graduation – May 2027
GPA: 3.73/4.00
Major: Computer Science
Minors: Bioinformatics, Integrated Biology
President's Academic Honors: Fall 2023 – Present

RESEARCH EXPERIENCE

Biotechnology Research & Development Intern — May 2026 – August 2026
Corteva Agriscience – Genome Center of Excellence — Johnston, Iowa

- Engineered a full-stack Python and Streamlit application to automate OMNI genomic probe panel tracking, replacing a manual spreadsheet-based process that was a documented bottleneck for the R&D team's sample throughput
- Integrated the application with internal genomic databases via REST APIs and implemented JWT-based authentication (AEGIS), enabling secure, real-time data exchange between R&D and bioinformatics systems
- Built a custom email-parsing workaround using the win32com library to extract structured shipment data from Outlook distribution lists, circumventing Microsoft Graph API limitations and eliminating a manual data-entry step in the logistics workflow
- Collaborated with infrastructure engineering on Azure DevOps CI/CD pipeline design and Bicep-based deployment, improving the reliability and maintainability of research software as it moved toward production use

Computational Biology Research Intern — March 2026 – Present
Yale School of Medicine — Remote
PI – Dr. Ruslan Medzhitov and Dr. Allison Greaney
Project: Investigating Lung Biology through Tissue-Engineered Bioreactor Models

- Analyze single-cell RNA sequencing (scRNA-seq) data using Seurat in R to characterize gene expression patterns in lung tissue models
- Perform scRNA-seq analysis and visualization in R using Seurat, ggplot2, and related bioinformatics package
- Perform end-to-end scRNA-seq analysis pipeline, from raw data quality control and normalization through dimensionality reduction, cell clustering, and differential expression, to characterize transcriptional differences between BCL5 and BCL6 bioreactor conditions in lung tissue models

Research Assistant — August 2025 – Present

Madison Cyber Labs — Madison, South Dakota
PI – Dr. Justin Blessinger
Cognitive Accessibility & Multisensory Environment Design

- Designed a Snoezelen multisensory environment using CAD tools, human-centered design frameworks, and accessibility standards for users with cognitive and sensory disabilities
- Developed sensory interaction systems integrating programmable lighting, audio controls, and tactile stimuli with embedded hardware coding

Assistive Hearing & Audio Signal Processing

- Conducted quantitative audio signal analysis using Python and MATLAB to evaluate amplification, latency, and noise reduction
- Performed comparative benchmarking of accessibility features across consumer earbud platforms

Assistive Biomedical Device Prototyping

- Designed and fabricated wearable assistive device prototypes through iterative prototyping cycles using Prusa and LulzBot TAZ 3D printers, evaluating form, fit, and usability with target user populations
- Applied rapid prototyping workflows and makerspace fabrication tools to develop low-cost, accessible hardware solutions for users with disabilities

AI-Based Accessibility Tools

- Conducted systematic evaluation of AI voice recording tools using speech enhancement and noise suppression algorithms
- Benchmarked transcription accuracy and usability using Python-based analysis workflows

Undergraduate Ecology Researcher — August 2025 – March 2026
Dakota State University Department of Arts & Sciences — Madison, South Dakota
PI – Dr. Kristel Bakker

- Conducted field-based survey of Great Blue Heron rookeries in South Dakota, quantifying colony metrics and comparing findings against three decades of prior literature (Baker et al. 2015, Naugle et al. 1996, Dowd 1982)
- Analyzed spatial and habitat data to assess landscape-level ecological change across survey sites
- Investigated neonicotinoid contamination in wild bird eggshells, detecting imidacloprid residues across multiple species using laboratory analysis and microscopy — potentially first field-based detection of this breadth in the region; manuscript in preparation

Research Intern — May 2025 – August 2025
University of Utah Department of Mechanical Engineering — Salt Lake City, Utah
Project: High Throughput Experimentation and Data Analysis Revealing the Intricate Interplay of Oil Spills and Microplastics Across Diverse Conditions
PI – Dr. Samira Shiri

- Selected as one of 25 scholars in a highly competitive NIH-funded Summer Program for Undergraduate Research
- Designed and executed a high-throughput experimental workflow to investigate particle-surface interactions, generating large-scale image-based datasets for quantitative analysis

- Developed an automated image analysis pipeline using ImageJ and MATLAB to quantify dynamic physical phenomena of polyethylene, polystyrene and oils
- Developed structured workflows using C++ and Excel to improve efficiency, transparency, and reporting accuracy
- Presented findings and safety-focused recommendations to faculty using PowerPoint and LaTeX
- Presented findings in poster presentation at University of Utah Summer 2025 Annual Symposium

Undergraduate Biology Researcher August 2024 – May 2025
Dakota State University Department of Arts & Sciences *Madison, South Dakota*
Project: Survey of Prairie Lakes for Microplastics in Eastern South Dakota
PI – Dr. Kristel Bakker

- Conducted field sampling across multiple prairie lakes to assess microplastic contamination in freshwater ecosystems
- Performed laboratory-based analysis to identify and classify microplastics by morphology using microscopy
- Applied light microscopy to characterize samples and distinguish contamination types across lake sites
- Presented findings at National Conference on Undergraduate Research (NCUR) 2025

Software Engineering Fellow June 2024 – July 2024
HeadStarter AI Remote

- Engineered Python-based data processing and automation tools to streamline research workflows
- Performed quantitative validation and data quality checks to maintain accuracy and audit readiness
- Delivered reports and visualizations supporting data-driven decision-making
- Collaborated with stakeholders to translate technical findings into process improvements

TECHNICAL SKILLS

Programming: Python, R, MATLAB, C++, SQL
Bioinformatics: Seurat, scRNA-seq analysis, differential expression, transcriptomics
Data Analysis & Visualization: ggplot2, ComplexHeatmap, ImageJ, Excel
Hardware & Prototyping: CAD, 3D printing, embedded systems
Laboratory: Microscopy, field sampling, experimental workflows

LEADERSHIP EXPERIENCE

Editor-in-Chief December 2023 – Present
Trojan Times – Dakota State University Newspaper *Madison, South Dakota*

- Lead a team of writers, editors, and designers to produce the university's flagship publication
- Designed and conducted an in-depth interview with the Governor of South Dakota, translating complex policy discussions into accessible, fact-checked reporting
- Oversee the editorial process from pitch to publication, enforcing quality and accuracy standards
- Develop content strategy engaging the campus community on science, research, and technology

AWARDS AND HONORS

- Student Research Initiative Award (2×) – Dakota State University
- Second Best Poster Presentation – Dakota State University
- President's Academic Honors List

- Dakota State University Rising Scholar
- Dakota State University Champion Scholar

SELECTED ABSTRACTS & CONFERENCE PRESENTATIONS

Oral Presentations

- **Ayelazono, B**. (2026, April). Nest Success and Habitat Characteristics of a First-Year Great Blue Heron Rookery. Presented at the International Association for Landscape Ecology – North America (IALE-NA). Athens, GA.
- **Ayelazono, B**. (2025, April). Survey of Prairie Lakes for Microplastics in Eastern South Dakota. Presented at Dakota State University Research Seminar, Madison, SD.

Poster Presentations

- **Ayelazono, B**. (2026, April). Detection of Neonicotinoid in Eggshells of Wild Bird Species. Presented at South Dakota Academy of Science Annual Symposium, Madison, SD.
- **Ayelazono, B**. (2025, July). High Throughput Experimentation and Data Analysis Revealing the Intricate Interplay of Oil Spills and Microplastics Across Diverse Conditions. Presented at the University of Utah Summer Conference. Salt Lake City, UT.
- **Ayelazono, B**. (2025, April). Survey of Prairie Lakes for Microplastics in Eastern South Dakota. Presented at the National Conference of Undergraduate Research, Pittsburgh, PA.
- **Ayelazono, B.** (2025, March). Survey of Prairie Lakes for Microplastics in Eastern South Dakota. Presented at Dakota State University Annual Research Symposium. Madison, SD.

Manuscript in Preparation

- **Ayelazono, B**.; Bakker, Kristel. Detection of Neonicotinoid in Eggshells of Wild Bird Species across South Dakota. Target journal TBD.

Published Abstract

- **Ayelazono, B**.; Shiri, Samira. High Throughput Experimentation and Data Analysis Revealing the Intricate Interplay of Oil Spills and Microplastics Across Diverse Conditions. *Range* 26:12 (2025). https://uen.pressbooks.pub/range26i2/chapter/ayelazono/